

$$\rho_{in} = |0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \xrightarrow{\tilde{G}^d} \begin{pmatrix} 0 & 0 \\ 0 & (2p-1)^d \end{pmatrix}$$
$$\langle \tilde{Z} \rangle = \text{Tr}(\tilde{Z} \rho_{out}) = \langle Z \rangle_{ideal} = (1)^d \quad \text{if } p=0 \Rightarrow \langle Z \rangle_{ideal} = (1)^d$$
$$\langle \tilde{Z} \rangle = (2p-1)^d \approx 1 - 2d \overset{\text{error}}{p} \quad \text{Linear in } p$$

